

Identity Operators

Q1: What are identity operators in Python? Explain their usage with examples.

A1:

Identity operators in Python are used to compare the memory location of two objects to check if they refer to the same object.

is: This operator returns True if both variables refer to the same object in memory.

is not: This operator returns True if both variables do not refer to the same object.

Q2: How do identity operators differ from equality operators?

A2:

Equality operators (== and !=) check if the values of two variables are the same.

Identity operators (is and is not) check if the two variables refer to the same object in memory.

For example:

python

```
a = [1, 2, 3]
```

```
b = a
```

```
c = [1, 2, 3]
```

```
print(a == c) # True, because the values are the same
```

```
print(a is c) # False, because they are different objects in memory
```

Membership Operators

Q3: What are membership operators in Python? Provide examples of how they are used.

A3:

Membership operators are used to test if a value is a member of a sequence such as a list, tuple, or string.

in: Returns True if the value is found in the sequence.

not in: Returns True if the value is not found in the sequence.

Example:

python

```
list_example = [1, 2, 3, 4, 5]
```

```
print(3 in list_example) # True, because 3 is an element of the list
```

```
print(6 not in list_example) # True, because 6 is not an element of the list
```

Q4: How would you check if a specific substring exists within a string using membership operators?

A4:

You can use the in operator to check if a substring exists within a string. If the substring is present, the expression will return True; otherwise, it will return False.

Example:
python

```
text = "Hello, world!"
```

```
print("world" in text) # True, because "world" is a substring of "Hello, world!"  
print("Python" not in text) # True, because "Python" is not a substring of "Hello, world!"
```

Output Formatting (F-strings)

Q5: What are F-strings in Python and why are they preferred over other formatting methods?

A5:

F-strings, introduced in Python 3.6, are a way to embed expressions inside string literals using curly braces {}. They are preferred because they are more readable, concise, and faster than older methods like % formatting or str.format().

Example:
python

```
name = "John"  
age = 25
```

```
print(f"My name is {name} and I am {age} years old.")
```

Q6: How can you perform calculations within an F-string? Provide an example.

A6:

You can directly perform calculations or call functions within an F-string by placing the expression inside the curly braces {}.

Example:
python

```
a = 5  
b = 10
```

```
print(f"The sum of {a} and {b} is {a + b}.") # The sum of 5 and 10 is 15.
```